

Population Analysis Process

Data Collection and Spatial Analysis

The *tidycensus* package in R was used to collect 2024 total population data for Iowa counties, including county geometry. The data were then exported as a shapefile using the `st_write` function. The shapefile was imported into Tableau to visualize the spatial distribution of population across Iowa counties and to gain a clearer understanding of county-level population patterns.

The 2024 population figures were sorted in descending order to identify the five most populous counties in Iowa. The shapefile was also imported into GeoDa to conduct Exploratory Spatial Data Analysis (ESDA). This analysis helped identify clusters of counties with high populations surrounded by counties with similarly high populations, as well as counties with low populations surrounded by counties with low populations. Two spatial weight matrices were applied in the analysis: the Queen Contiguity matrix and the K-Nearest Neighbor matrix. The outputs from the ESDA were exported as images for use in the final Tableau dashboard.

Following this step, total population data for the five identified counties were collected for the years 2013–2024 from the ACS 5-year estimates. The datasets were cleaned and organized in Excel to prepare them for analysis and visualization.

Components of Population Change Analysis

Research was conducted to obtain county-level birth and death statistics for the five selected counties. However, comprehensive county-level birth and death data were not available for all years required for the study. As an alternative, county-level migration data were obtained from the IRS migration dataset, which provided migration estimates through 2022. These datasets were also cleaned and organized in Excel for analysis.

Using the available population and migration data, Natural Increase was calculated using the formula:

$$\text{Natural Increase} = \text{Population Change} - \text{Net Migration}$$

This calculation assumes that overall population change is equal to the sum of net migration and natural increase. The resulting dataset provided estimates of the major components of population change for the selected counties.

Final Visualization and Dashboard Development

ArcGIS was used to create a shapefile containing the five selected counties for use in dashboard development. Population growth trends for the five counties were visualized in Tableau using line graphs, while Net Migration and Natural Increase were displayed using column charts.

These visualizations were combined into interactive dashboards showing population distribution across Iowa, the identification of the top five most populous counties, and the population change

analysis for the selected counties. Finally, the dashboards were compiled into a Tableau Story, published to Tableau Public, and embedded into the project's GitHub web portfolio.